

Transverse divergences of the p^+ -integrated row

UV, collinear, soft — scanned on $2a + 2b + 2c + 2d + 2e + 2f$

$$A = (x - z)^2, B = (x - w)^2, C = (z - w)^2, A' = (x' - z)^2, B' = (x' - w')^2, C' = (z - w')^2; \mathcal{K}_{ww'} \equiv \frac{(x - w) \cdot (x' - w')}{(x - w)^2 (x' - w')^2}$$

The six parts are unchanged and still exact, and both fixes landed: $2f$ now carries $\int_{z,w,w',x,x'}$, and the N_c -collapsed colour bracket is used throughout (§1). **On divergences: there is exactly one, and it is soft.** No UV and no collinear log anywhere — all 26 short-distance regions are convergent *after* the p^+ integration as well as before (§2). The single surviving transverse log is $|z| \rightarrow \infty$, and its coefficient is

$$\frac{\mathcal{K}_{ww'}}{4k^+} \log \frac{\vee - k^+}{\Lambda},$$

matched to 7–8 digits at four choices of $(\Lambda, \vee)$ (§3). Two colour index-ordering issues remain, and one of them now decides whether your N_c exists at all (§5).

1. The six parts

Every coefficient is character-for-character what I checked last time, so the verification stands: each reproduces its p^+ master to 10^{-16} – 10^{-12} relative. Both corrections are in. One cosmetic leftover: $2f$'s **first piece still carries the unexpanded product** of the two colour brackets while its other three pieces carry the expanded four-term version. They are equal, so nothing is wrong — but make it uniform.

2. Short distance: no UV, no collinear

A log needs the integrand $\sim \rho^{-2(n-1)}$ when n points collapse, since the measure is $\rho^{2(n-1)} \frac{d\rho}{\rho}$. Scanning all 26 subsets of $\{x, x', z, w, w'\}$ on the p^+ -integrated bracket:

no short-distance divergence in any of the 26 regions.

The physically named ones, in detail:

region	what it is	behaviour	against measure	verdict
$z \rightarrow w$	the two final gluons at one point — final-state collinear	ρ^{-1} $\langle G \rangle \rho = 4.6352$ flat over $10^{-2} \dots 10^{-6}$	$\rho d\rho$	convergent , no collinear log
$w \rightarrow x$	observed gluon on its source	ρ^{-1} (4.9948)	$\rho d\rho$	convergent — the ordinary $1/r$ of the WW kernel
$z \rightarrow x$	unobserved gluon on the source	ρ^{-1} (1.5891)	$\rho d\rho$	convergent
$w' \rightarrow w$	the $k_\perp$ -integrated corner	$O(1)$ (0.7671)	$\rho d\rho$	regular
$\{z, w, w'\}$	both emission points on the unobserved gluon	ρ^{-2} (50-digit)	$\rho^3 d\rho$	convergent

Why there is no collinear log, restated for the integrated expression. At $z \rightarrow w$ only *one* of the two vertices is singular: the conjugate sits at $w' \neq w$ and is regular there, and $e^{-ik_\perp(w'-w)}$ is what keeps w' from following w . One singular kernel gives ρ^{-1} ; a log needs ρ^{-2} , i.e. both. Forcing both by sending $w' \rightarrow w$ too does give ρ^{-2} , but then the measure is $\rho^3 d\rho$ and you are further from divergence, not closer. The p^+ integration does not change this: it multiplies each region by a bounded factor.

3. Large distance: one soft log, and that is all

variable $\rightarrow \infty$	$R = 10^2$	10^3	10^4	verdict
z	6.31637	6.31557	6.31557	log divergent
x	3.96586	3.96592	3.98058	log divergent in the bare kernel
x'	2.83707	2.83713	2.83769	log divergent in the bare kernel
w, w'	oscillatory, does not settle			Fresnel — conditionally convergent

Entries are $|\langle G \rangle_{\hat{n}}| R^2$; a constant means a $\int d^2 R / R^2 \log$.

3.1 The $|z| \rightarrow \infty$ log — the one real divergence

Λ	$\vee$	measured $\langle G \rangle z ^2$	$\frac{\mathcal{K}_{ww'}}{4k^+} \log \frac{\vee - k^+}{\Lambda}$
10^{-4}	10^5	4.7366752	4.7366752
10^{-6}	10^5	5.7892702	5.7892702
10^{-6}	10^8	7.3681650	7.3681650
10^{-8}	10^6	7.3681647	7.3681647

The two soft regions are one region. $|z| \rightarrow \infty$ is the unobserved gluon's transverse momentum going to zero; $p^+ \rightarrow \Lambda$ is its longitudinal momentum going to zero. They are not independent divergences — the transverse log's coefficient is the rapidity span $\log \frac{\vee - k^+}{\Lambda}$, and the transverse structure left over, $\mathcal{K}_{ww'}$, is exactly the LO single-gluon-production structure. The z -gluon has decoupled completely. That is the eikonal region, and it is what the JIMWLK/BK subtraction removes. In your decomposition it is $2b$ and $2c$ that produce it: their $\log |z|^2$ pieces cancel against each other and the $\frac{B+B'}{B'-B} \log \frac{B'}{B}$ leftovers cancel against $2a$.

3.2 The $|x|, |x'| \rightarrow \infty$ tails are not loop divergences

x and x' are the arguments of ρ . The colour-charge correlator has the transverse support of the target, so these ends are cut at the target radius in the standard way. Nothing to regulate.

4. The taxonomy, answered directly

type	where it would live	present?
UV	points collapsing — any of 26 subsets	No. Every region ρ^{-1} or milder.
Collinear (final state)	$z \rightarrow w$	No. ρ^{-1} , not ρ^{-2} .
Collinear (initial state)	$w \rightarrow x, z \rightarrow x$	No. ρ^{-1} — the WW kernel's integrable $1/r$.
Soft (transverse)	$ z \rightarrow \infty$	Yes — one log, coefficient above.
Soft (longitudinal)	$p^+ \rightarrow \Lambda$, i.e. part $2b$	Yes — the same divergence, seen from the other side.
Power	anywhere	No. $2f$'s $[\frac{1}{\vee} - \frac{1}{k^+ + \Lambda}] \rightarrow -\frac{1}{k^+}$.

So this row needs no dimensional regularisation and no $\overline{\text{MS}}$ subtraction. It needs exactly one thing: the rapidity subtraction. Its $\int^R \frac{d^2 z}{|z|^2}$ closes against N_1 , the $(2\rho) \times (2\rho)$ piece and the virtual rows.

5. Colour: two different index-ordering problems

5.1 The z -side decides whether you get N_c at all

The collapse runs through $\sum_b [\text{bra } U(z)]^{bc'} [\text{ket } U(z)]^{bc} = \delta^{c'c}$. That needs the **same** index order on both sides:

bra / ket at z	$\ \mathcal{C}_{22} - N_c[\cdot]\ $	result
both $U(z)$ (your six parts)	$9 \cdot 10^{-15}$	collapses
both $U^\dagger(z)$	$5 \cdot 10^{-15}$	collapses
mixed (your top line)	2.58	does not collapse

Because $\sum_b U^{bc'} U^{bc} = \delta^{c'c}$ and $\sum_b U^{c'b} U^{cb} = \delta^{c'c}$, but $\sum_b U^{bc'} U^{cb} = [UU]^{cc'}$ is not a delta. Your **top-line** expression writes $U^{\dagger bc}(z)$ in the ket and $U^{bc'}(z)$ in the bra — that combination gives no N_c . Your expanded parts use $U^{bc}(z)$ on both sides, which does. Fix the top line to match; either convention works, as long as it is the same one twice.

5.2 The x -side decides which N_c structure

bra's x' line	gives	check
$U^{e'd}(x')$ (as written)	$N_c[U(x')U(x)]^{e'e}$	$9 \cdot 10^{-15}$
$U^{d'e'}(x')$ (conjugate of the ket)	$N_c[U^\dagger(x')U(x)]^{e'e}$	$5 \cdot 10^{-15}$

The two structures differ by 4.03 in norm. The second is the adjoint **dipole** and is what you want physically.

Numerical warning, restated because it bit the scan. The $\{z, w, w'\}$ collapse drives $AB' - A'B \rightarrow 0$, and float64 loses the cancellation below $\rho \sim 10^{-3}$: it reported $1.2 \cdot 10^{21}$ where the true value is $3.46 \cdot 10^{12}$. The $|x| \rightarrow 10^5$ entry above is degraded for the same reason. Any numerical integration of these six parts must expand analytically near $A = B$, $A' = B'$ and $AB' = A'B$, or work in extended precision there.

Checked by `src/row20_div.py`, `src/row20_quant.py`, `src/row20_colour.py`; output in `RESULTS_row20.txt`.